

Workflow Audit & AI Integration Blueprint

Phase: Setup — AI Fluency & Foundations

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Role: AI/ML Intern @ FlyRank AI

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1. Weekly Workflow Audit (12 Tasks)

Classification of recurring weekly tasks based on human-in-the-loop requirements, risk profile, and automation potential.

#	Recurring Task	Classification	One-Line Rationale
1	Writing custom loss functions for ML models	Just me	Requires deep mathematical logic; AI hallucinations can introduce silent bugs into gradient descent.
2	Reviewing and approving PRs from peers	Just me	Code ownership, production safety, and peer feedback require genuine human accountability and context.
3	Drafting initial Python code for data pipelines	Collaborate with AI	Accelerates boilerplate creation while allowing me to steer edge-case handling and schema design.
4	Generating unit tests for preprocessing functions	Delegate with review	AI excels at exhaustive test case generation, but I must verify coverage and boundary conditions.
5	Debugging PyTorch / scikit-learn error traces	Collaborate with AI	Speeds up pinpointing obscure dimension mismatch errors while I evaluate and apply the fix.
6	Formatting pandas EDA tables into Markdown	Fully automate	Rigid structural task with zero creative necessity; seamlessly handled via simple prompts or scripts.
7	Summarizing daily research papers on arXiv	Delegate with review	Extracts core methodology quickly, but I need to scan key figures and formulas for accuracy.
8	Writing weekly internship status reports	Collaborate with AI	Converts raw bullet points into polished updates while retaining my authentic engineering voice.
9	Organizing and tagging daily notes in Notion	Fully automate	Pure administrative overhead that automated rules or scripts handle without human intervention.
10	Drafting documentation for pipeline scripts	Delegate with review	Generates inline docstrings and high-level summaries that I audit before merging into main.
11	Learning new ML concepts (Transformer math)	Collaborate with AI	Using AI as an interactive Socratic tutor allows me to test my understanding in real time.
12	Searching for baseline datasets on Hugging Face	Just me	Selecting correct data distribution requires nuanced project context and strict license checks.

2. Free Toolkit & Claude Project Setup

- [✓] **Claude Account:** Configured with Custom Project Instructions
- [✓] **ChatGPT Account:** Verified & Active
- [✓] **Anthropic Academy:** Enrolled in *AI Fluency: Framework & Foundations* (Module 1 Completed)

Configured Claude Project Custom Instructions

The following custom prompt is set up in the "**FlyRank ML Internship**" Claude Project:

You are an expert AI/ML Peer and Engineering Collaborator assisting Shahzaib Pervez, an AI/ML Intern at FlyRank AI.

Persona & Tone:

- Direct, technical, concise, and incisive. Zero fluff or conversational filler.
- High-density technical communication calibrated for a machine learning engineer.

Preferences:

- Prioritize clean Python code using standard libraries (pandas, PyTorch, scikit-learn).
- When debugging or writing code, explain the 'why' behind architectural choices briefly.
- Adhere to modular, readable code standards suitable for production git repositories.

Current Goals:

- Building readable ML pipelines, mastering model evaluation on real search data, and completing the FlyRank AI internship curriculum.

3. Target Reuse Tasks & Measurable Success Definitions

These three tasks will be reused in assignments FL-02 through FL-04:

Task 1: Generating Unit Tests for Data Preprocessing Pipelines

- **Classification:** Delegate to AI with review
- **Measurable Success Definition:** Generates 100% syntactically correct `pytest` functions covering null values, schema shifts, and data type mismatches. Achieves >90% code coverage on target script without introducing false positives.

Task 2: Summarizing ML Research Papers & Technical Docs

- **Classification:** Delegate to AI with review
- **Measurable Success Definition:** Produces a structured 3-bullet executive summary (Problem, Novel Architecture, Results/Metrics). Accurately identifies practical implementation tradeoffs without hallucinating figures.

Task 3: Converting Raw EDA Observations into Structured Markdown Reports

- **Classification:** Collaborate with AI
- **Measurable Success Definition:** Transforms unformatted pandas console outputs into clean, scannable Markdown tables. Generates 3 actionable insights on model performance/data distributions with 0% data error.

Submission Instructions

1. Save this document into your local workspace as `work/phase_setup_workflow_audit.pdf`.
2. Add a screenshot of your configured Claude Project interface inside your submission folder.
3. Commit and push to your public repo: `git commit -m "docs: complete phase setup workflow audit"`
4. Submit your repo URL: <https://github.com/Shahzaib-Pervez/flyrank-ml-internship>